

Carter Rennick

Toronto, ON, Canada | carter.rennick@gmail.com | (613) 329-8544 | gamesbycarter.vercel.app
linkedin.com/in/carterrennick | github.com/ShikenNuggets

Professional Summary

Generalist software developer with 6+ years of experience in game development and real-time 3D applications, with a focus on C++, proprietary game engines, and Unreal Engine 5. Proficient in procedural and AI-driven automation, gameplay programming, telemetry, and low-level subsystems such as skeletal animation and rendering. Proven experience collaborating with cross-discipline teams, optimizing production pipelines, and eliminating technical debt.

Experience

Unreal Engine Software Developer, Clay Software Corporation Oct 2024 – present

- Consolidated disparate prototype functionality from Unreal Blueprints into organized, performant, and production-ready C++ plugins, allowing new features and bug fixes to be seamlessly rolled out to dozens of ArchViz projects.
- Built automation workflows with C++-based Unreal Editor tools, Python scripts, and MCP tools to generate 3D environments from client-provided 2D schematics, significantly reducing setup time for new projects.
- Designed a self-serve customization system to allow lighting, materials, and furniture placement to be configured at runtime without needing to rebuild/redeploy applications.
- Created a highly intuitive player navigation system for exploring the digital twins, requiring minimal setup per-project and providing an accessible, easy to learn experience for non-technical users.
- Optimized RAM and GPU usage, and reduced worst-case load times by 80% to ensure smooth operation in our AWS cloud environment.

Programmer, Gameloft Toronto July 2019 – Apr 2024

- Championed C++ data tracking integration for [Disney Magic Kingdoms](#)' iOS version, collaborating with the Features team to measure usage of major gameplay and e-commerce features with minimal runtime overhead.
- Eliminated redundant and unnecessary data to optimize storage costs while still meeting our data clients' needs.
- Analyzed decade old legacy code to identify and resolve long-standing data tracking issues.
- Integrated and updated external libraries using Premake.
- Ported data tracking, UI, and anti-tampering code from a proprietary C++ game engine to Unity/C#.

Peer Tutor, Humber College Oct 2018 – Apr 2019

- Ran workshops and one-on-one tutoring sessions to help students understand math and programming concepts.

Education

Humber College, Game Programming (Advanced Diploma) 2016 – 2019

- Capstone Project: led a team of students in creating a custom C++ game engine and a [physics-based 3D platformer](#).

Projects

GadgetEngine | C++ / OpenGL 2022 – present

- Developed a [3D-capable game engine](#) and [reusable C++ component library](#), featuring a custom OpenGL renderer, skeletal animation, and platform abstractions on top of SDL3.
- Used the engine to [publish a game](#) for the GMTK Game Jam in 2024.

Secret Identity | UE5 / C++ Apr 2024 – June 2024

- Created a vertical slice for a [superhero action game](#) in Unreal Engine 5.4, implementing flight mechanics with cape physics, randomized enemy encounters, and UMG-based UI.

Skills

Programming Languages: C++, C#, Python

Game Engines: Unreal Engine 5.3 - 5.8, Proprietary C++ Engines, Unity

Graphics and Math: Linear Algebra, OpenGL, Skeletal Animation, Shaders (GLSL/HLSL)

Tooling: CMake, Premake, Git, SVN, Visual Studio, Vscope